Calculation of Median for Grouped Data or Continuous Series

Example 13.4

Calculate the median for the following data

Literates by age in 2001

Age group	Literacy in %
5 – 10	22.3
10 – 15	81.7
15 – 20	79.3
20 – 25	73.2
30 – 35	64.5
35 – 40	50

Solution:

Age group	Frequency f	Cumulative Frequency c f	
5 – 10	22.3	22.3	
10 – 15	81.7	104	
15 – 20	79.3	183.3	
20 – 25	**73.2**	**256.5**	→ Median class
30 – 35	64.5	321	
35 – 40	50	371	
	N = 371		

Calculation of Median Class

$$N = 371, \left(\frac{N}{2}\right) = \left(\frac{371}{2}\right) = 185.5$$

Class corresponding to 185.5 is 20-25 i.e. the Median Class

Formula

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

where

l = Lower limit of the median class
N = Total Numbers of frequeny
f = Frequeny of the median class
m = Cumulative frequency of the class preceding the median class
c = The class interval of the median class.

TN_GOVT_XII_Geography_chapter13.indd 200 12/6/2022 6:05:02 PM